

FMDBS — Access DB vs SRS Gap Analysis

Analysis by Antigravity | June 2026 Sources: MLE Records NHK_be (1).accdb · PM Records NHK_be (1).accdb · FMDBS_SRS_Tira-miss-u.pdf v1.0

1. Overview of What Exists

The repository contains two **split-database** Microsoft Access applications:

File Pair	Purpose	Tables
MLE Records NHK MAC.accdb (frontend)	UI / forms / queries only	0 user tables
MLE Records NHK_be.accdb (backend)	Actual data store	7 tables
PM Records NHK MAC.accdb (frontend)	UI / forms / queries only	0 user tables
PM Records NHK_be.accdb (backend)	Actual data store	10 tables

Both frontends list named queries that act as named views: Claim Summary, Contact Log, Documents, MLE Records, Photos, Reports, Samples (MLE); Claims Summary, Contact Log, Documents, Investigations PM, PM Records, Photos, Reports, tblCODAbbreviations, tblCODRules, tblUsers (PM).

2. Extracted Schema Summary

2.1 MLE (Clinical) Backend — 7 Tables

MLE Records — Core clinical case table (36 fields)

Field	Type	Notes
ID	Integer	Auto PK
CME_No	Text(255)	Primary case reference
MLEF_No	Text(255)	Police-issued MLEF number
BHT_No	Text(250)	Bed Head Ticket
Full_Name	Text(255)	Patient name
Address	Text(250)	

Sex	Text(6)	
Age_Years	Text(255)	Stored as text, not numeric
Examined_date	Date/Time	NOT NULL
Category	Text(50)	Case type
Date_of_Trial	Date/Time	Court date
Courts	Text(255)	
Police	Text(255)	Police station
Summoned	Text(255)	Summons reference
Place	Text(255)	Place of examination
Age_Days	Text(255)	
Produced_BY	Text(255)	Who produced the patient
Seen_BY	Text(255)	Examining doctor
Age_Months	Text(255)	
Evidence	Text(255)	
Case_No	Text(255)	
MLEF_Status	Text(255)	
Contact_No	Text(255)	Patient contact
Evidence_Claim	Text(255)	
Completed_ON	Date/Time	
SR_No	Text(255)	Serial/register number
Contact_No_Police	Text(255)	
Tags	Memo	Free-text tags
MLRGeneratedOn	Date/Time	
TRClaim_Date	Date/Time	Travel claim date
TRClaim_Amount	Currency	Travel claim amount
Summon_Received_ON	Date/Time	
Hurt	Memo	Free-text injury description
TOXGeneratedOn	Date/Time	Toxicology report generated
STIGeneratedOn	Date/Time	STI report generated

PSYCHGeneratedOn	Date/Time	Psychiatry report generated
Contact Log — Communications log (8 fields)		
CME_No, Reference, Contents (Memo), Actions (Memo), Date_Log, Full_Name, Contact_No		
Documents — Scanned document file references (5 fields)		
CME_No, FilePath, FileName, FileType		
Photos — Photo file references (5 fields)		
CME_No, FilePath, FileName, FileType		
Reports — Report dispatch tracking (12 fields)		
CME_No, Full_Name, Reports_TO, Reports_Via, Reports_Date, Reports_Certificate, MLEF_No, RC_No, Claim_Reports, Claim_SentON, Claim_ReceivedON		
Samples — Investigation/specimen table (9 fields)		
CME_No, Full_Name, Date_Sampled, Specimen, Sent_TO, Results, MLEF_No, Ix_No		
Claim Summary — Financial claims table (8 fields)		
High_Courts, Type, Amount (Currency), Sent_On, Received_On, Claim_Status, Voucher		

2.2 PM (Autopsy) Backend — 10 Tables

PM Records — Core autopsy case table (66 fields)

Key fields beyond MLE equivalents: | Field | Type | Notes | |-----|-----|-----| | PM_No | Text(255) | Primary autopsy reference | | Case_No | Text(50) | Inquest/court case number | | Morgue_No | Text(255) | | Name_of_Deceased | Text(255) | | COD | Text(250) | Cause of death (primary) | | COD_1b/1c/1d | Text(255) | WHO antecedent/contributory causes | | COD_2 | Text(255) | Other significant conditions | | Verdict | Text(255) | Accident/suicide/natural/open | | Done_BY | Text(255) | Examining doctor | | Date_of_MorAdm | Date/Time | Morgue admission date | | Time_of_MorAdm | Date/Time | Morgue admission time | | Date_of_Adm | Date/Time | Hospital admission | | Time_of_Adm | Date/Time | | Date_of_Death | Date/Time | | Time_of_Death | Date/Time | | PHN_No | Text(255) | Preliminary hospital number | | NOK | Text(255) | Next of Kin | | Relationship | Text(255) | | NIC | Text(255) | Deceased NIC | | Date_BP | Date/Time | Body release date | | Time_BP | Date/Time | | Tech_BP | Text(255) | Technician for body release | | DC_No | Text(255) | Death certificate number | | Tech_1, Tech_2 | Text(255) | Assisting technicians | | Cemetery | Memo | | Mode | Text(255) | Mode of death | | Storage | Text(255) | Body storage details | | DD_No | Text(255) | | Religion | Text(255) | | Police_Officer | Text(255) | | Evidence/Evidence_Claim | Text(255) | | TrialCourts/TrialCase_No | Text(255) | Trial court details | | CODGeneratedOn, TOXGeneratedOn, PMRGeneratedOn, HISTOGeneratedOn, MICROGeneratedOn | Date/Time | Report generation timestamps |

Investigations PM — Specimens and lab investigations (13 fields)

Extends MLE Samples: adds Discarded_On, Date_Results, Specimen_No (Integer), CT_No

tblCODAbbreviations — COD abbreviation lookup

Abbreviation, FullText, IsActive, Area
tblCODRules — Automated COD template-filling rules
Phrase, TargetBookmark, InsertText, AppendMode, Priority, IsActive
tblUsers — User authentication table
UserID, Username, Password (Text 255), DoctorName, IsAdmin (Boolean), IsActive (Boolean)
Contact Log, Documents, Photos, Reports, Claims Summary
Equivalent to MLE counterparts; Reports uses PM_No as foreign key, Contact Log adds Contact_Person field, Claims Summary has Cheque_No instead of Voucher.

3.1 Gap: What the Access DB Has That the SRS Lacks or Under-specifies

3.1 Financial / Claims Management (Entire domain missing from SRS)

The Access DB has a full **claims management subsystem** absent from the SRS:

Access DB Field	Meaning
TRClaim_Date, TRClaim_Amount	Travel reimbursement claims made by doctors to courts
Claim Summary table	Tracks claims sent to High Courts with Type, Amount, Sent_On, Received_On, Claim_Status, Voucher/Cheque_No
Reports.Claim_Reports, Claim_SentON, Claim_ReceivedON	Claim lifecycle tracking per report
Claims Summary.Claim_Status	Whether a claim was paid/pending/rejected

SRS Improvement needed: The SRS explicitly states the system "will NOT manage financial transactions" — but the real-world workflow **does** involve court fee claims by forensic doctors. The SRS should either formally include a lightweight **Court Claim Tracking** module, or at minimum acknowledge this as a future consideration. The existing Access DB proves this is an active, daily-use feature.

3.2 Contact Log / Communication History

The Access DB has a dedicated Contact Log table for both MLE and PM:

Field	Description
Reference	Communication reference

Contents (Memo)	Full text of communication
Actions (Memo)	Actions taken
Date_Log (auto-timestamped)	When logged
Full_Name , Contact_No	Who contacted

SRS Gap: FR-NOTIF covers outgoing notifications only. There is no requirement for **incoming communication logging** — i.e., recording phone calls, walk-ins, or correspondence received from families, police, or courts about a case. The Access DB shows this is a real operational need. The SRS should add a **FR-COMM** requirement for a case-linked contact/communication log.

3.3 Age Stored as Three Separate Text Fields

The Access DB stores age as **Age_Years** (Text!), **Age_Months** (Text), **Age_Days** (Text) — all as Text (255).

SRS Gap: Appendix B lists **Age** as a single field. The SRS should specify: - Age shall be stored as Years/Months/Days (three integer fields) — because forensic age estimation of children often requires sub-year precision. - Age should be **Integer**, not Text. The Access DB's use of Text creates data quality risks (e.g., "3 years" vs "3" vs "3yrs" all valid).

3.4 Produced_BY / Seen_BY Fields (Chain of Custody)

The MLE Records table tracks: - **Produced_BY** — who brought the patient (police officer?) - **Seen_BY** — which doctor examined them

SRS Gap: FR-CLIN-01 captures the examining doctor but does not specify tracking **who produced/brought the patient** (typically a police officer escorting the person). This is a chain-of-custody detail with legal significance. The SRS should capture this as a required field under FR-CLIN-01.

3.5 Multiple Report Generation Timestamps

Both DB backends track **separate generation timestamps per report type**: **MLRGeneratedOn**, **TOXGeneratedOn**, **STIGeneratedOn**, **PSYCHGeneratedOn** (MLE) - **CODGeneratedOn**, **TOXGeneratedOn**, **PMRGeneratedOn**, **HISTOGeneratedOn**, **MICROGeneratedOn** (PM)

SRS Improvement: The SRS treats report generation monolithically (FR-CLIN-04: "Generate Report"). The real system needs to independently track when each report type (Toxicology, Histology, Psychiatry, STI, MLR, COD, PMR, Microbiology) was generated. The SRS should define each report type as a distinct tracked document with its own **GeneratedOn**, **DispatchedOn**, and **ReceivedOn** timestamps.

3.6 Specimen / Investigation Discarding

The `Investigations` PM table has a `Discarded_On` field — indicating specimens are formally discarded after a period.

SRS Gap: There is no requirement covering **specimen lifecycle management** — i.e., logging when a specimen is sent, when results come back (`Date_Results`), and when the specimen is formally discarded. This should be added under FR-AUTO-05.

3.7 Next-of-Kin (NOK) and Body Release (PM only)

The PM Records table has a cluster of body release fields completely absent from the SRS:

Field	Meaning
<code>NOK_Relationship</code>	Next of kin details
<code>NIC</code>	Deceased's NIC
<code>Date_BR</code> , <code>Time_BR</code>	Body release date/time
<code>Tech_BR</code>	Technician who released the body
<code>DC_No</code>	Death certificate number
<code>Cemetery</code>	Where buried/cremated
<code>Mode</code>	Mode of death delivery context
<code>Storage</code> , <code>DD_No</code>	Body storage and DD number
<code>Religion</code>	For religious rites at release

SRS Gap: FR-AUTO-01 (Autopsy Case Registration) does not include **body release workflow** — which is a major operational post-autopsy step. The SRS should add a `FR-AUTO-09: Body Release and Next-of-Kin Management` requirement covering: NOK details, body release authorization, Death Certificate number, storage, and religious considerations.

3.8 Magistrate Court vs High Court Distinction

`Claim_Summary.High_Courts` implies claims are specifically sent to High Courts, while `MLE_Records.Courts` tracks the trial court. The PM table has both `Courts` (inquest) and `TrialCourts` (subsequent trial).

SRS Gap: FR-CLIN-01 lists "Court" as a single field. The SRS should distinguish between: 1. **Ordering court** (magistrate court that issued the MLEF/inquest order) 2. **Trial court** (where the doctor is eventually summoned to testify — may be a High Court)

3.9 COD Rules Engine (Autopsy)

4.4 Referral Status Lifecycle

UC-02 defines referral states: **Pending** → **Completed** with document upload triggering state change and read-only lock.

The Access DB `Samples` table has no `status` field — only `Results`. There is no referral-specific table in the MLE DB (referrals appear to be tracked via `Samples/Contact Log` informally).

A proper `Referrals` table is needed with: `referral_type`, `referred_to_department`, `referred_doctor`, `referral_date`, `status`, `document_path`, `completed_on`.

4.5 Audio Recording and Transcript

FR-ADTC-04 and Product Function #4 require audio recording during FIM and transcript generation.

No audio recording, storage, or transcript field exists in either Access DB — this is an entirely new capability.

4.6 Notification System

No notification or scheduling tables exist in the Access DB. The SRS requires: - In-system notifications (FR-NOTIF-01) - SMS/WhatsApp notifications (FR-NOTIF-02) - Court date reminders (7-day, 3-day, 1-day)

New tables needed: `Notifications`, `NotificationConfig`, `SMSLog`.

4.7 Document Versioning

FR-DOC-03 requires version history for documents. The Access DB `Documents` table is a flat single-record-per-file store with no `version`, `previous_version_id`, or `uploaded_by` fields.

4.8 CME_No as Primary Key (Denormalization Risk)

The Access DB uses `CME_No` (Text) as the join key across all related tables — but it is not defined as a formal foreign key constraint in the schema (ADODB shows no referential integrity). It also appears to duplicate `MLEF_No` in some tables.

The SRS should clarify and standardize: use a surrogate integer PK internally, with `CME_No` / `MLEF_No` / `PM_No` as business-layer unique identifiers. The new PostgreSQL schema must enforce FK constraints.

5. Structural Recommendations for SRS

#	Recommendation	Priority
1	Add FR-CLAIM : Court claim tracking (doctor fee claims per case)	High — active operational need
2	Add FR-COMM : Case-linked contact/communication log	High — active operational need
3	Add FR-AUTO-09 : Body release and NOK management	High — active operational need
4	Clarify age storage as 3 integer fields (Years/Months/Days) in Appendix B	High
5	Add chain-of-custody field Produced_By to FR-CLIN-01	Medium
6	Expand FR-CLIN-04 / FR-AUTO-08 to list all report types with independent timestamps	Medium
7	Add FR-AUTO-05 specimen lifecycle (sent → results → discarded)	Medium
8	Distinguish ordering court vs trial court in FR-CLIN-01 / FR-AUTO-01	Medium
9	Describe COD rules/phrase engine in FR-AUTO-06	Medium
10	Add explicit note on credential migration from legacy plaintext Access DB	Medium
11	Add FR-AUTH-04 : Role table with 5 roles replacing binary IsAdmin flag	High
12	Add FR-RPT-03 audit log table schema to technical appendix	High
13	Add normalized Patient entity to data model (multi-case linkage)	High
14	Add Referrals table spec with status lifecycle to FR-CLIN-03	High
15	Add audio recording storage schema to FR-AUTO-04	Medium
16	Add document versioning schema to FR-DOC-03	Medium

6. Fields Present in Access DB Not in SRS Appendix B (MLEF Data

Fields)

The following fields exist operationally but are not listed in Appendix B of the SRS:

Field	DB Table	Missing from SRS
SR_No	MLE Records	Serial register number
Produced_BY	MLE Records	Who brought the patient
Age_Days, Age_Months	MLE Records	Sub-year age precision

